DBMS WEEKLY TASK 2

Identifying Data Requirements for an Online Retail System

About Primary Key
A Primary Key (PK) is a unique identifier assigned to every record in a database table. It helps the database distinguish one record from another and ensures that duplicate records are not created. A primary key cannot contain NULL values and is commonly used to establish relationships between multiple tables.

Example:
Every customer is assigned a unique Customer_ID such as CUS001. Even if two customers have the same name, their Customer_ID will always be different.

Main Data Entities
1. Customer: Customer_ID (PK), Customer_Name, Email, Phone_Number, Password, Address, Date_of_Birth.
2. Product: Product_ID (PK), Product_Name, Brand_Name, Price, Quantity_Available, Category_ID, Supplier_ID.
3. Category: Category_ID (PK), Category_Name, Description.
4. Seller: Seller_ID (PK), Seller_Name, Business_Name, Contact_Number, Email, Address.
5. Inventory: Inventory_ID (PK), Product_ID, Stock_Quantity, Last_Updated.
6. Orders: Order_ID (PK), Customer_ID, Order_Date, Order_Status, Total_Price.
7. Payment: Payment_ID (PK), Order_ID, Payment_Mode, Amount, Payment_Status.
8. Delivery: Delivery_ID (PK), Order_ID, Courier_Name, Tracking_ID, Delivery_Status.

Additional Data Requirements
Search History: Search_ID, Customer_ID, Search_Text, Search_Date.
Product Filters: Category, Brand, Price, Customer Rating, Discount, Availability, Color, Material.
Sorting Options: Best Match, Latest Products, Lowest Price, Highest Price, Most Popular.
Gender Category: Men, Women, Kids, Unisex.
Coupons: Coupon_ID, Coupon_Code, Discount_Value, Expiry_Date.
Return & Refund: Return_ID, Order_ID, Return_Reason, Refund_Status, Return_Date.

Conclusion
The online retail database is organized using well-defined entities and primary keys to ensure accurate data management, efficient relationships, secure transactions, and reliable inventory, order, payment, and delivery tracking.